

Math 502

Homework #2

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Exercise 2.2.

- (a) Determine the Green's function for the two-point boundary value problem $u''(x) = f(x)$ on $0 < x < 1$ with a Neumann boundary condition at $x = 0$ and a Dirichlet condition at $x = 1$; i.e., find the function $G(x; \bar{x})$ solving

$$G''(x; \bar{x}) = \delta(x - \bar{x}), \quad G'(0) = 0, \quad G(1) = 0,$$

the function $G_0(x)$ solving

$$G_0''(x) = 0, \quad G_0'(0) = 1, \quad G_0(1) = 0$$

and the function $G_1(x)$ solving

$$G_1''(x) = 0, \quad G_1'(0) = 0, \quad G_1(1) = 1.$$

- (b) Using this as guidance, find the general formulas for the elements of the inverse of the matrix in equation (2.54). Write out the 5×5 matrices of A and A^{-1} for the case $h = 0.25$.

Solution. (a) Since $\delta(x - \bar{x}) = 0$ whenever $x \neq \bar{x}$, $G''(x; \bar{x}) = 0$ whenever $x \neq \bar{x}$. Therefore:

$$G(x; \bar{x}) = \begin{cases} Ax + B, & x \in [0, \bar{x}) \\ Cx + D, & x \in [\bar{x}, 1] \end{cases}$$

Since Green's function is continuous, it must be the case that $A\bar{x} = B = C\bar{x} = D$. Note that Green's function has a jump condition at $\bar{x}$; i.e.:

$$\int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} G''(x, \bar{x}) dx = \int_{\bar{x}-\epsilon}^{\bar{x}+\epsilon} \delta(x - \bar{x}) dx = 1$$

Simplifying the left side gives $G'(\bar{x} + \epsilon; \bar{x}) - G'(\bar{x} - \epsilon; \bar{x}) = 1$; i.e., $C - A = 1$. Now our boundary conditions give that $A = 0$ and $C + D = 0$. From the 4 equations obtained, solving the system yields $A = 0$, $B = \bar{x} - 1$, $C = 1$, and $D = -1$. Thus:

$$G(x; \bar{x}) = \begin{cases} \bar{x} - 1, & x \in [0, \bar{x}) \\ x - 1, & x \in [\bar{x}, 1] \end{cases}$$

The functions $G_0(x)$ and $G_1(x)$ can be solved by simplifying integrating twice and using the boundary conditions to find the coefficients. We will find that $G_0(x) = x - 1$ and $G_1(x) = 1$.

(b)

$$A = \begin{pmatrix} -4 & 4 & 0 & 0 & 0 \\ 16 & -32 & 16 & 0 & 0 \\ 0 & 16 & -32 & 16 & 0 \\ 0 & 0 & 16 & -32 & 16 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -1 & -\frac{3}{16} & -\frac{1}{8} & -\frac{1}{16} & 1 \\ -\frac{3}{4} & -\frac{3}{16} & -\frac{1}{8} & -\frac{1}{16} & 1 \\ -\frac{1}{2} & -\frac{1}{8} & -\frac{1}{8} & -\frac{1}{16} & 1 \\ -\frac{1}{4} & -\frac{1}{16} & -\frac{1}{16} & -\frac{1}{16} & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Exercise 2.3. (Solvability Condition for Neumann Problem)

Determine the null space of the matrix A^T , where A is given in Equation (2.58), and verify that the condition (2.62) must hold for the linear system to have solutions.

Solution. Equation (2.58) from the textbook is:

$$\frac{1}{h^2} \begin{pmatrix} -h & h & & & \\ 1 & -2 & 1 & & \\ & 1 & -2 & 1 & \\ & & \ddots & \ddots & \ddots \\ & & & 1 & -2 & 1 \\ & & & & h & -h \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \\ u_2 \\ \vdots \\ u_m \\ u_{m+1} \end{pmatrix} = \begin{pmatrix} \sigma_0 + \frac{h}{2}f(x_0) \\ f(x_1) \\ f(x_2) \\ \vdots \\ f(x_m) \\ -\sigma_1 + \frac{h}{2}f(x_{m+1}) \end{pmatrix}$$

The transpose of the matrix A is:

$$A^T = \frac{1}{h^2} \begin{pmatrix} -h & 1 & & & \\ h & -2 & 1 & & \\ & 1 & -2 & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & -2 & h \\ & & & & 1 & -h \end{pmatrix}$$

To find the null space of A^T , we can row reduce the following augmented matrix:

$$\frac{1}{h^2} \left(\begin{array}{cccccc|c} -h & 1 & & & & 0 \\ h & -2 & 1 & & & 0 \\ & 1 & -2 & \ddots & & 0 \\ & & \ddots & \ddots & 1 & \vdots \\ & & & 1 & -2 & h & 0 \\ & & & & 1 & -h & 0 \end{array} \right).$$

After careful row operations, we get that the null space of A^T is $\text{span}\{(1, h, h, \dots, h, 1)^T\}$. Since a singular linear system has a solution if and only if the right-hand side is orthogonal to the null space of A^T , we can conclude that Equation (2.58) has a solution.

Exercise 2.4 prelim. Find the exact solution to the problem:

$$u''(x) = e^x, \quad x \in (0, 3)$$
$$u(0) = \alpha, \quad u'(3) = \sigma.$$

by simply integrating both sides of the differential equation as many times as necessary. Use this exact solution to test the code in Exercise 2.4. Try a couple of different values of α and σ for yourself, but choose only one set of values to submit as part of your homework solution.

Solution. The solution to this differential equation given the boundary conditions is $u(x) = e^x + (\sigma - e^3)x + (\alpha - 1)$. The values that I will submit for this assignment will be $\sigma = e^3$ and $\alpha = 1$.

Exercise 2.4.

- (a) Modify the m-file `bvp_2.m` so that it implements a Dirichlet boundary condition at $x = a$ and a Neumann condition at $x = b$ and test the modified program.
- (b) Make the same modification to the m-file `bvp_4.m`, which implements a fourth order accurate method. Again test the modified program.

Solution. (a) Here is the adjusted code and a log-log output:

```
1 % bvp_2.m
2 % second order finite difference method for the bvp
3 % u''(x) = f(x), u(ax)=beta, u'(bx)=sigma
4 % Using 3-pt differences on an arbitrary nonuniform grid.
5 % Should be 2nd order accurate if grid points vary smoothly,
   but may
6 % degenerate to "first order" on random or nonsmooth grids.
7 % Different BCs can be specified by changing the first and/or
   last rows of
8 % A and F.
9 ax = 0;
10 bx = 3;
11 sigma = exp(3); % Neumann boundary condition at bx
12 beta = 1; % Dirichlet boundary condition at ax
13 f = @(x) exp(x); % right hand side function
14 utrue = @(x) exp(x) + (sigma-exp(bx))*(x - ax) + beta - exp(
   ax); % true soln
15 % true solution on fine grid for plotting:
16 xfine = linspace(ax,bx,101);
17 ufine = utrue(xfine);
18 % Solve the problem for ntest different grid sizes to test
   convergence:
```

```

19 m1vals = [10 20 40 80 160 1000];
20 ntest = length(m1vals);
21 hvals = zeros(ntest,1); % to hold h values
22 E = zeros(ntest,1); % to hold errors
23 for jtest=1:ntest
24     m1 = m1vals(jtest);
25     m2 = m1 + 1;
26     m = m1 - 1; % number of interior grid points
27     hvals(jtest) = (bx-ax)/m1; % average grid spacing, for
        convergence tests
28 % set grid points:
29 gridchoice = 'uniform'; % see xgrid.m for other choices
30 x = xgrid(ax,bx,m,gridchoice);
31 % set up matrix A (using sparse matrix storage):
32 A = spalloc(m2,m2,3*m2); % initialize to zero matrix
33 % first row for Dirichlet BC at ax:
34 A(1,1:3) = fdcoeffF(0, x(1), x(1:3));
35 % interior rows:
36 for i=2:m1
37     A(i,i-1:i+1) = fdcoeffF(2, x(i), x((i-1):(i+1)));
38 end
39 % last row for Neumann BC at bx:
40 A(m2,m:m2) = fdcoeffF(1, x(m2), x(m:m2));
41 % Right hand side:
42 F = f(x);
43 F(1) = beta;
44 F(m2) = sigma;
45 % solve linear system:
46 U = A\F;
47 % compute error at grid points:
48 uhat = utrue(x);
49 err = U - uhat;
50 E(jtest) = max(abs(err));
51 disp(' ')
52 disp(sprintf('Error with %i points is %9.5e',m2,E(jtest)))
53 clf
54 plot(x,U,'o') % plot computed solution
55 title(sprintf('Computed solution with %i grid points',m2));
56 hold on
57 plot(xfine,ufine) % plot true solution
58 hold off

```

```

59 % pause to see this plot:
60 drawnow
61 input('Hit <return> to continue ');
62 end
63 error_table(hvals, E); % print tables of errors and ratios
64 error_loglog(hvals, E); % produce log

```

```
>> untitled2
```

```
Error with 11 points is 1.74886e+00
Hit <return> to continue
```

```
Error with 21 points is 4.80811e-01
Hit <return> to continue
```

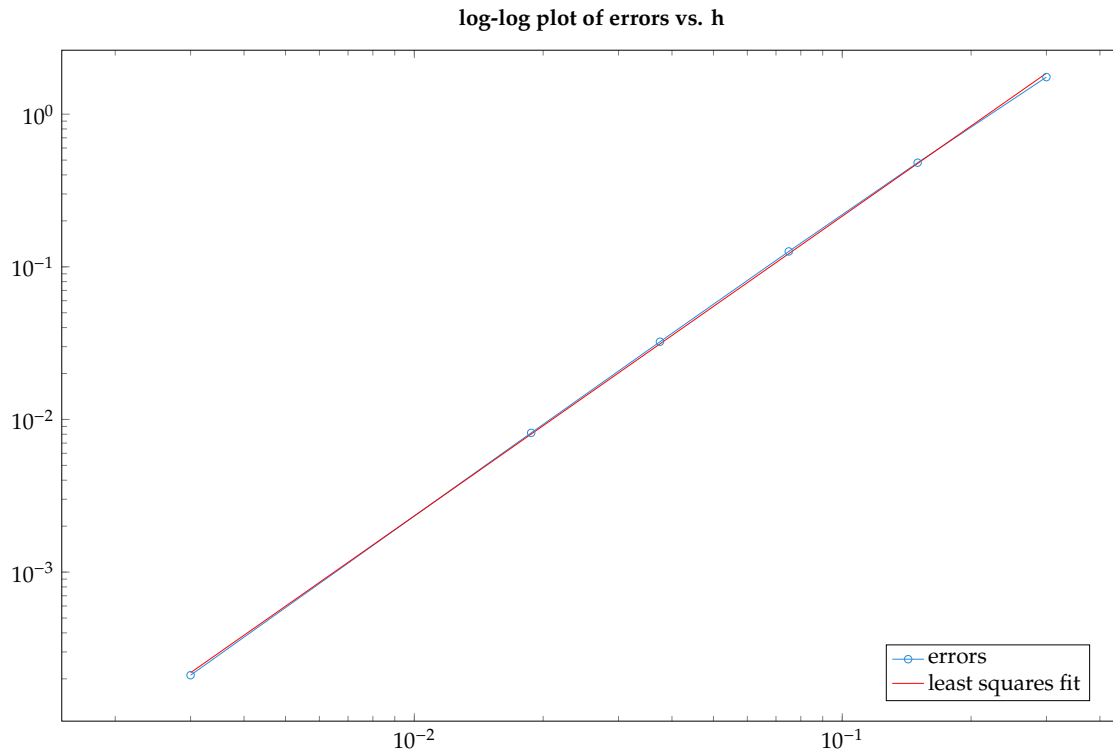
```
Error with 41 points is 1.26086e-01
Hit <return> to continue
```

```
Error with 81 points is 3.22858e-02
Hit <return> to continue
```

```
Error with 161 points is 8.16884e-03
Hit <return> to continue
```

```
Error with 1001 points is 2.11242e-04
Hit <return> to continue
```

h	error	ratio	observed order
0.30000	1.74886e+00	NaN	NaN
0.15000	4.80811e-01	3.63732	1.86288
0.07500	1.26086e-01	3.81336	1.93106
0.03750	3.22858e-02	3.90531	1.96544
0.01875	8.16884e-03	3.95231	1.98270
0.00300	2.11242e-04	38.67058	1.99450



(b) Here is the adjusted code and a log-log output:

```

1 % bvp4.m
2 % second order finite difference method for the bvp
3 % u''(x) = f(x), u(ax)=beta, u'(bx)=sigma
4 % fourth order finite difference method for the bvp
5 % u'' = f, u(ax)=beta, u'(bx)=sigma
6 % Using 5-pt differences on an arbitrary grid.
7 % Should be 4th order accurate if grid points vary smoothly.
8 % Different BCs can be specified by changing the first and/or
9 % last rows of
10 % A and F.
11 ax = 0;
12 bx = 3;
13 sigma = exp(3); % Neumann boundary condition at bx
14 beta = 1; % Dirichlet boundary condition at ax
15 f = @(x) exp(x); % right hand side function
16 utrue = @(x) exp(x) + (sigma-exp(bx))*(x - ax) + beta - exp(ax); % true soln
17 % true solution on fine grid for plotting:
18 xfine = linspace(ax,bx,101);
19 ufine = utrue(xfine);

```

```

19 % Solve the problem for ntest different grid sizes to test
    convergence:
20 m1vals = [10 20 40 80];
21 ntest = length(m1vals);
22 hvals = zeros(ntest,1); % to hold h values
23 E = zeros(ntest,1); % to hold errors
24 for jtest=1:ntest
25     m1 = m1vals(jtest);
26     m2 = m1 + 1;
27     m = m1 - 1; % number of interior grid points
28     hvals(jtest) = (bx-ax)/m1; % average grid spacing, for
        convergence tests
29 % set grid points:
30 gridchoice = 'uniform';
31 x = xgrid(ax,bx,m,gridchoice);
32 % set up matrix A (using sparse matrix storage):
33 A = spalloc(m2,m2,5*m2); % initialize to zero matrix
34 % first row for Dirichlet BC on u(x(1))
35 A(1,1:5) = fdcoeffF(0, x(1), x(1:5));
36 % second row for u'(x(2))
37 A(2,1:6) = fdcoeffF(2, x(2), x(1:6));
38 % interior rows:
39 for i=3:m
40     A(i,i-2:i+2) = fdcoeffF(2, x(i), x((i-2):(i+2)));
41 end
42 % next to last row for u'(x(m+1))
43 A(m1,m-3:m2) = fdcoeffF(2,x(m1),x(m-3:m2));
44 % last row for Neumann BC on u'(x(m+2))
45 A(m2,m-2:m2) = fdcoeffF(1,x(m2),x(m-2:m2));
46 % Right hand side:
47 F = f(x);
48 F(1) = beta;
49 F(m2) = sigma;
50 % solve linear system:
51 U = A\F;
52 % compute error at grid points:
53 uhat = utrue(x);
54 err = U - uhat;
55 E(jtest) = max(abs(err));
56 disp(' ')
57 disp(sprintf('Error with %i points is %9.5e',m2,E(jtest)))

```

```

58 clf
59 plot(x,U,'o') % plot computed solution
60 title(sprintf('Computed solution with %i grid points',m2));
61 hold on
62 plot(xfine,ufine) % plot true solution
63 hold off
64 % pause to see this plot:
65 drawnow
66 input('Hit <return> for next plot ');
67 end
68 error_table(hvals, E); % print tables of errors and ratios
69 error_loglog(hvals, E); % produce log

```

```
>> untitled3
```

```
Error with 11 points is 7.22610e-02
```

```
Hit <return> for next plot
```

```
Error with 21 points is 5.21206e-03
```

```
Hit <return> for next plot
```

```
Error with 41 points is 3.47061e-04
```

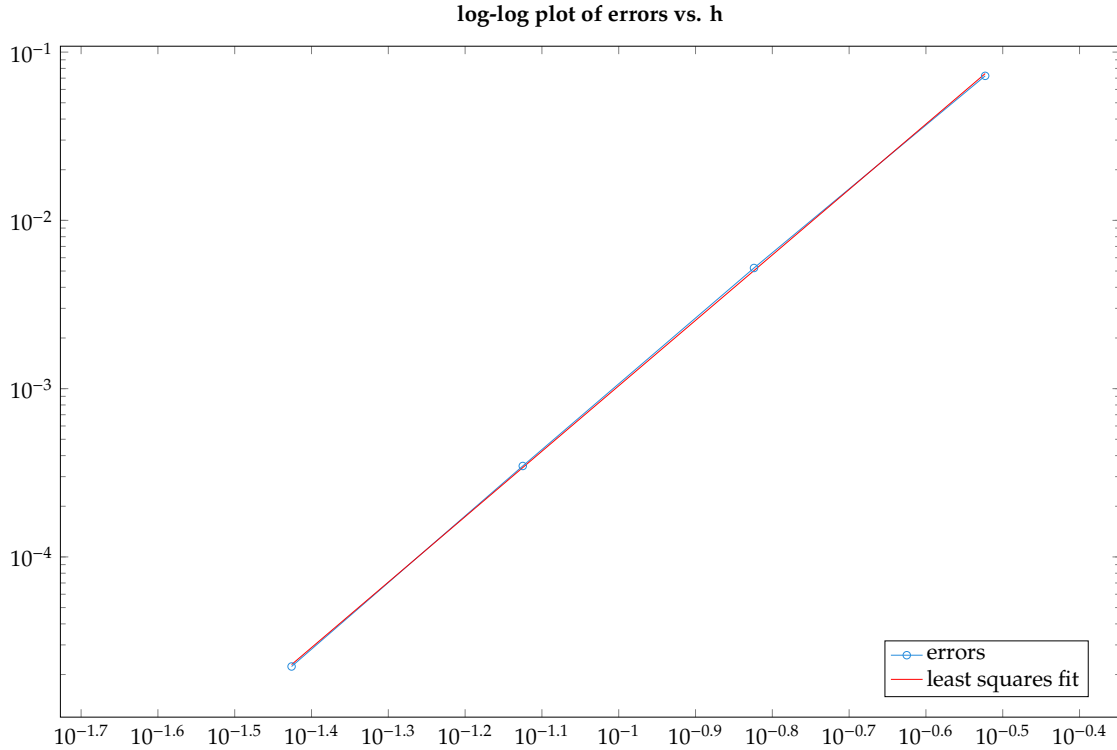
```
Hit <return> for next plot
```

```
Error with 81 points is 2.23260e-05
```

```
Hit <return> for next plot
```

h	error	ratio	observed order
0.30000	7.22610e-02	NaN	NaN
0.15000	5.21206e-03	13.86421	3.79329
0.07500	3.47061e-04	15.01768	3.90859
0.03750	2.23260e-05	15.54515	3.95839

```
Least squares fit gives  $E(h) = 8.04933 * h^{3.88894}$ 
```

Exercise 2.6. Consider the following linear boundary value problem with Dirichlet boundary conditions:

$$\begin{aligned} u''(x) &= u(x) = 0, \quad x \in (a, b) \\ u(a) &= \alpha, \quad u(b) = \beta. \end{aligned}$$

Note that this equation arises from a linearized pendulum.

- Modify the m-file `bvp_2.m` to solve this problem. Test your modified routine on the problem with $a = 0$, $b = 1$, $\alpha = 2$, $\beta = 3$. Determine the exact solution for comparison.
- Let $a = 0$ and $b = \pi$. For what values of α and β does this boundary value problem have solutions? Sketch a family of solutions in a case where there are infinitely many solutions
- Solve the problem with $a = 0$, $b = \pi$, $\beta = -1$ using your modified `bvp_2.m`. Which solution to the boundary value problem does this appear to converge to as $h \rightarrow 0$? Change the boundary value at $b = \pi$ to $\beta = 1$. Now how does the numerical solution behave as $h \rightarrow 0$.
- You might expect the linear system in (c) to be singular since the boundary value problem is not well posed. It is not, because of discretization error. Compute the eigenvalues of the matrix A for this problem and show that an eigenvalue approaches 0 as $h \rightarrow 0$. Also show that $\|A^{-1}\|_2$ blows up as $h \rightarrow 0$ so that the discretization is unstable.

Solution. (a) Here is the modified code and output:

```

1 % bvp_2.m
2 % second order finite difference method for the bvp
3 %  $u''(x) + u(x) = 0$ ,  $u(ax)=\alpha$ ,  $u(bx)=\beta$ 
4 % Using 3-pt differences on an arbitrary nonuniform grid.
5 % Should be 2nd order accurate if grid points vary smoothly,
   but may
6 % degenerate to "first order" on random or nonsmooth grids.
7 %
8 % Different BCs can be specified by changing the first and/or
   last rows of
9 % A and F.
10 %
11 % From http://www.amath.washington.edu/~rjl/fdmbook/ (2007)
12 ax = 0;
13 bx = 1;
14 alpha = 2; % Dirichlet boundary condition at ax
15 beta = 3; % Dirichlet boundary condition at bx
16 f = @(x) 0*x; % right hand side function
17 utrue = @(x) alpha*cos(x) + (beta-alpha*cos(bx))*sin(x)/sin(
   bx); % true soln
18 % true solution on fine grid for plotting:
19 xfine = linspace(ax,bx,101);
20 ufine = utrue(xfine);
21 % Solve the problem for ntest different grid sizes to test
   convergence:
22 m1vals = [10 20 40 80 160 1000];
23 ntest = length(m1vals);
24 hvals = zeros(ntest,1); % to hold h values
25 E = zeros(ntest,1); % to hold errors
26 for jtest=1:ntest
27     m1 = m1vals(jtest);
28     m2 = m1 + 1;
29     m = m1 - 1; % number of interior grid points
30     hvals(jtest) = (bx-ax)/m1; % average grid spacing, for
   convergence tests
31 % set grid points:
32 gridchoice = 'uniform'; % see xgrid.m for other choices
33 x = xgrid(ax,bx,m,gridchoice);
34 % set up matrix A (using sparse matrix storage):
35 A = spalloc(m2,m2,3*m2); % initialize to zero matrix

```

```

36 % first row for Dirichlet BC at ax:
37 A(1,1:3) = fdcoeffF(0, x(1), x(1:3));
38 % interior rows:
39 for i=2:m1
40 A(i,i-1:i+1) = fdcoeffF(2, x(i), x((i-1):(i+1)));
41 A(i,i) = A(i,i) + 1;
42 end
43 % last row for Dirichlet BC at bx:
44 A(m2,m:m2) = fdcoeffF(0,x(m2),x(m:m2));
45 % Right hand side:
46 F = f(x);
47 F(1) = alpha;
48 F(m2) = beta;
49 % solve linear system:
50 U = A\F;
51 % compute error at grid points:
52 uhat = utrue(x);
53 err = U - uhat;
54 E(jtest) = max(abs(err));
55 disp(' ')
56 disp(sprintf('Error with %i points is %9.5e',m2,E(jtest)))
57 clf
58 plot(x,U,'o') % plot computed solution
59 title(sprintf('Computed solution with %i grid points',m2));
60 hold on
61 plot(xfine,ufine) % plot true solution
62 hold off
63 % pause to see this plot:
64 drawnow
65 input('Hit <return> to continue ');
66 end
67 error_table(hvals, E); % print tables of errors and ratios
68 error_loglog(hvals, E); % produce log

```

```
>> problem2point6
```

```
Error with 11 points is 3.24687e-04
```

```
Hit <return> to continue
```

```
Error with 21 points is 8.10848e-05
```

```
Hit <return> to continue
```

```
Error with 41 points is 2.02705e-05
```

```
Hit <return> to continue
```

```
Error with 81 points is 5.06999e-06
```

```
Hit <return> to continue
```

```
Error with 161 points is 1.26748e-06
```

```
Hit <return> to continue
```

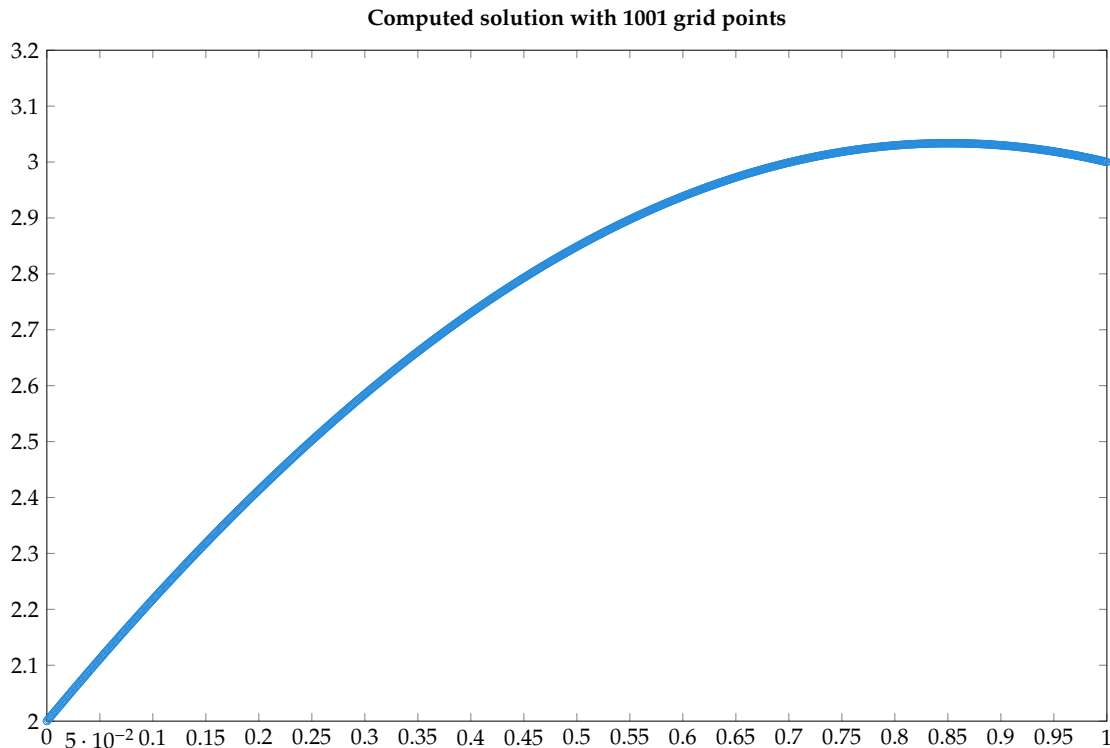
```
Error with 1001 points is 3.24062e-08
```

```
Hit <return> to continue
```

h	error	ratio	observed order
0.10000	3.24687e-04	NaN	NaN
0.05000	8.10848e-05	4.00430	2.00155
0.02500	2.02705e-05	4.00014	2.00005
0.01250	5.06999e-06	3.99812	1.99932
0.00625	1.26748e-06	4.00007	2.00002
0.00100	3.24062e-08	39.11215	2.00069

```
Least squares fit gives  $E(h) = 0.0324794 * h^{2.00029}$ 
```

```
>>
```



The exact solution for $a = 0$, $b = 1$, $\alpha = 2$, and $\beta = 3$ is $u(x) = 2 \cos(x) + \frac{3-2 \cos(1)}{\sin(1)} \sin(x)$.

(b) The boundary value problem has a solution if and only if $\beta = -\alpha$. There are infinitely many solutions

(c) Observe the following code, output, and plot.

```

1 % bvp_2.m
2 % second order finite difference method for the bvp
3 % u''(x) + u(x) = 0, u(ax)=alpha, u(bx)=beta
4 % Using 3-pt differences on an arbitrary nonuniform grid.
5 % Should be 2nd order accurate if grid points vary smoothly,
   but may
6 % degenerate to "first order" on random or nonsmooth grids.
7 %
8 % Different BCs can be specified by changing the first and/or
   last rows of
9 % A and F.
10 %
11 % From http://www.amath.washington.edu/~rjl/fdmbook/ (2007)
12 ax = 0;
13 bx = pi;
14 alpha = 1;
15 beta = -1; % first run
16 f = @(x) 0*x; % right hand side function

```

```

17 utrue = @(x) cos(x); % for comparison to the converged
    solution % true soln
18 % true solution on fine grid for plotting:
19 xfine = linspace(ax,bx,101);
20 ufine = utrue(xfine);
21 % Solve the problem for ntest different grid sizes to test
    convergence:
22 m1vals = [10 20 40 80 160 1000];
23 ntest = length(m1vals);
24 hvals = zeros(ntest,1); % to hold h values
25 E = zeros(ntest,1); % to hold errors
26 for jtest=1:ntest
27     m1 = m1vals(jtest);
28     m2 = m1 + 1;
29     m = m1 - 1; % number of interior grid points
30     hvals(jtest) = (bx-ax)/m1; % average grid spacing, for
        convergence tests
31 % set grid points:
32 gridchoice = 'uniform'; % see xgrid.m for other choices
33 x = xgrid(ax,bx,m,gridchoice);
34 % set up matrix A (using sparse matrix storage):
35 A = spalloc(m2,m2,3*m2); % initialize to zero matrix
36 % first row for Dirichlet BC at ax:
37 A(1,1:3) = fdcoeffF(0, x(1), x(1:3));
38 % interior rows:
39 for i=2:m1
40     A(i,i-1:i+1) = fdcoeffF(2, x(i), x((i-1):(i+1)));
41     A(i,i) = A(i,i) + 1;
42 end
43 % last row for Dirichlet BC at bx:
44 A(m2,m:m2) = fdcoeffF(0,x(m2),x(m:m2));
45 % Right hand side:
46 F = f(x);
47 F(1) = alpha;
48 F(m2) = beta;
49 % solve linear system:
50 U = A\F;
51 % compute error at grid points:
52 uhat = utrue(x);
53 err = U - uhat;
54 E(jtest) = max(abs(err));

```

```

55 disp(' ')
56 disp(sprintf('Error with %i points is %9.5e',m2,E(jtest)))
57 clf
58 plot(x,U,'o') % plot computed solution
59 title(sprintf('Computed solution with %i grid points',m2));
60 hold on
61 plot(xfine,ufine) % plot true solution
62 hold off
63 % pause to see this plot:
64 drawnow
65 input('Hit <return> to continue ');
66 end
67 error_table(hvals, E); % print tables of errors and ratios
68 matlab2tikz('myfile2.tex');
69 error_loglog(hvals, E); % produce log

```

```
>> problem2point6
```

```
Error with 11 points is 2.31489e-03
```

```
Hit <return> to continue
```

```
Error with 21 points is 5.73244e-04
```

```
Hit <return> to continue
```

```
Error with 41 points is 1.44354e-04
```

```
Hit <return> to continue
```

```
Error with 81 points is 3.60616e-05
```

```
Hit <return> to continue
```

```
Error with 161 points is 9.01416e-06
```

```
Hit <return> to continue
```

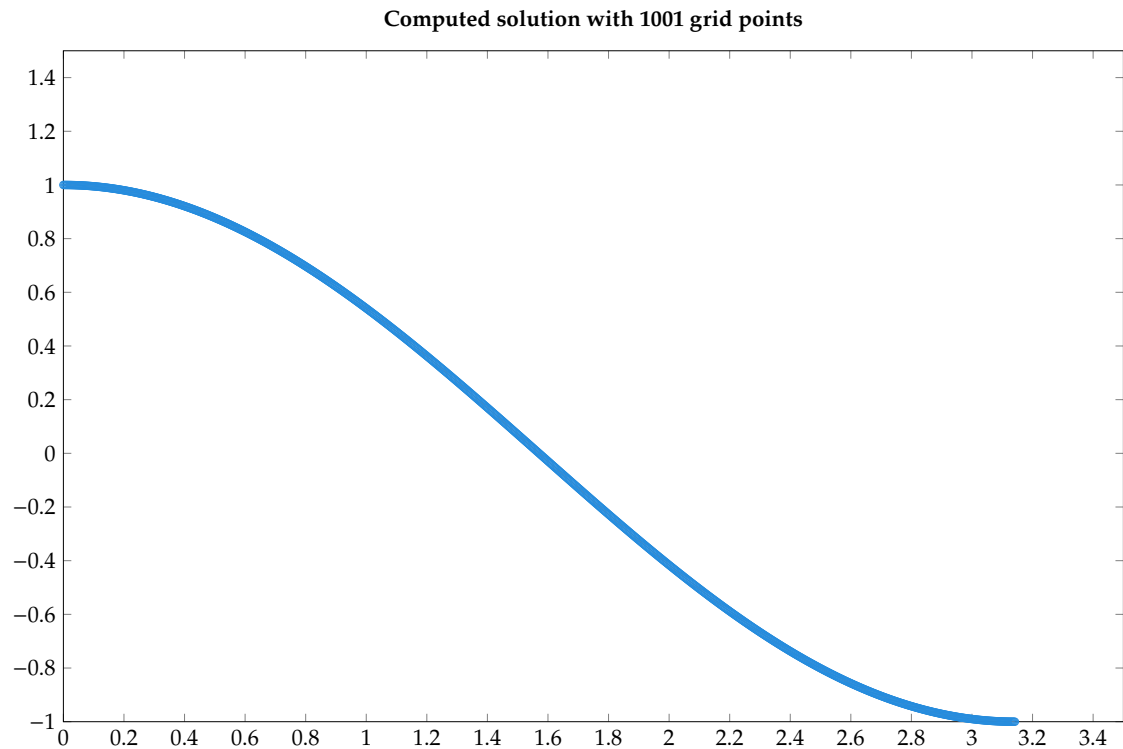
```
Error with 1001 points is 7.74535e-06
```

```
Hit <return> to continue
```

h	error	ratio	observed order
0.31416	2.31489e-03	NaN	NaN
0.15708	5.73244e-04	4.03823	2.01372
0.07854	1.44354e-04	3.97111	1.98954
0.03927	3.60616e-05	4.00297	2.00107
0.01963	9.01416e-06	4.00055	2.00020

```
0.00314    7.74535e-06    1.16382    0.08278
```

Least squares fit gives $E(h) = 0.00483763 * h^{1.29861}$



We can see that this solution converges to $u(x) = \cos(x)$. Making the minimal change of $\beta = 1$ gives the following output and plot:

```
>> problem2point6

Error with 11 points is 1.63312e+16
Hit <return> to continue

Error with 21 points is 1.63312e+16
Hit <return> to continue

Error with 41 points is 1.63312e+16
Hit <return> to continue

Error with 81 points is 1.63312e+16
Hit <return> to continue
```



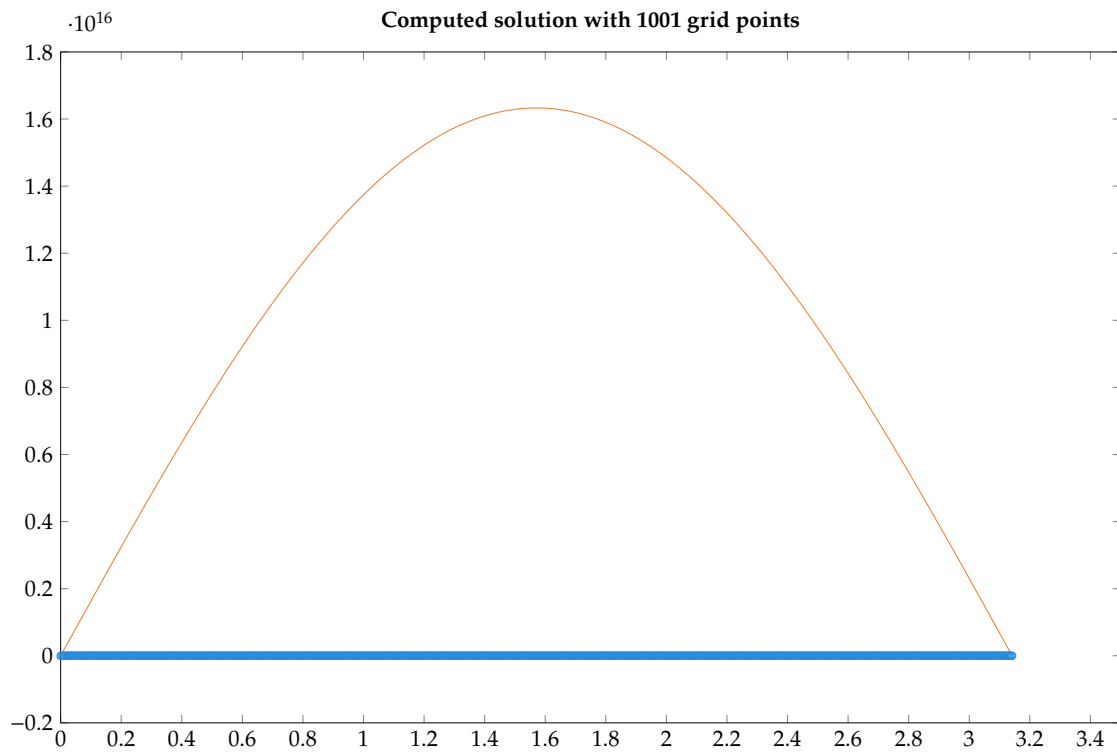
```
Error with 161 points is 1.63312e+16
Hit <return> to continue
```

```
Error with 1001 points is 1.63312e+16
Hit <return> to continue
```

h	error	ratio	observed order
0.31416	1.63312e+16	NaN	NaN
0.15708	1.63312e+16	1.00000	-0.00000
0.07854	1.63312e+16	1.00000	-0.00000
0.03927	1.63312e+16	1.00000	-0.00000
0.01963	1.63312e+16	1.00000	-0.00000
0.00314	1.63312e+16	1.00000	-0.00000

```
Least squares fit gives  $E(h) = 1.63312e+16 * h^{-1.9068e-11}$ 
```

```
>>
```



Since solvability requires $\beta = -\alpha$, the discretized solution does not converge to the true solution.

(d)